

Espen Haukeland Kristensen

Manager

espen.h.kristensen@pwc.com
95848889

Profile

Translation to English

Espen is a data engineer and AI developer at PwC with expertise in data platforms, generative AI, and cloud-based infrastructure on Azure and Databricks. He holds certifications as a Power BI Data Analyst (PL-300) and Microsoft Azure Data Fundamentals, and has a background in language technology, machine learning, mathematics, and computer science from the University of Oslo.

He has contributed to an Azure- and Databricks-based data platform for the Norwegian Environment Agency with independent delivery responsibility, and developed ChatNH – a secure internal AI chat application based on Azure AI Foundry – for IKT Nordhordland, with responsibility for infrastructure as code and security architecture.

Espen is analytical, collaborative, and passionate about his field, and communicates technical topics effectively to both technical and non-technical audiences.

- Key highlights:**

Data platforms (Azure, Databricks, medallion architecture)
Generative AI and large language model-based solutions (Azure AI Foundry)
ETL development in Python with CI/CD
Infrastructure as code (Bicep, Terraform)
Data visualisation and Power BI (certified PL-300)
Data security, GDPR, and the EU AI Act

Certifications

- PL-300 Microsoft Certified: Power BI Data Analyst Associate
- DP-900 Azure Data Fundamentals
- Confluent Fundamentals Accreditation
- IBM Certified Solution Developer – App Connect Enterprise V11

Work Experience

08.2025 - ongoing
PwC Norway
Manager

12.2022 - 07.2025
Kantega AS
Data Engineer

09.2021 - 11.2022
Avella AS
Integration Consultant

Education

2022
University of Oslo
Single courses (master level): language technology and machine learning

08.2018 - 01.2020
University of Oslo
Enkeltemner (master): språkteknologi og maskinlæring

08.2014 - 06.2018
University of Oslo
Bachelor Mathematics, Informatics and Technology

Project Experience

Duration: 11.2025 - ongoing **Scope:** 80&

Data Engineer / Data Analyst

Duration: 08.2025 - ongoing **Scope:** 70&

Norsk Pensjon

The Pension Account Register – Weekly Statistics Reporting

Duration: 09.2024 - 12.2024 **Scope:** 100&

Project description: Norsk Pensjon is a service providing Norwegians with an overview of their pension entitlements across various pension schemes. The company needed to automate the retrieval, storage, and accessibility of weekly statistics for the Pension Account Register, and sought to replace manual Excel reports with a more modern and interactive solution. The project delivered automated data pipelines and Power BI reports with expanded functionality, along with a flexible access control system. The solution reduced manual effort and provided decision-makers with faster and more reliable access to up-to-date statistics.

Role: Data Engineer
Role description: Espen established automated data pipelines for the retrieval and storage of weekly statistics from the Pension Account Register. He utilised Azure Databricks and Python for data processing, Terraform for infrastructure as code, and ETL processes to ensure structured and reliable data flow.

Role: Power BI Developer
Role description: Espen developed Power BI reports to replace the previous Excel reports, with expanded functionality and increased interactivity. He established a flexible access control system ensuring that the right information was available to the right users.

Norsk Pensjon

Secure Integration Provides Complete Pension Overview

Duration: 06.2024 - 09.2024 **Scope:** 100&

Project description: Norsk Pensjon is a service providing Norwegians with an overview of their pension entitlements across various pension schemes. The company needed to further develop its data platform in Azure into a new environment, as well as establish programmatic file uploads to the platform via API endpoints. The project delivered an expanded and modernised data platform that strengthened the company's capacity for statistical analysis and data processing. The solution provided increased flexibility and automation in data collection processes.

Role: Data Engineer
Role description: Espen further developed the data platform in Azure Synapse Analytics within a new Azure environment, and implemented programmatic file uploads via API endpoints. He utilised Bicep for infrastructure as code and Python together with Apache Spark for data processing, ensuring a smooth migration and expansion of the platform.

Sparebank 1 SMN

RAG and Norwegian Language Models

Duration: 05.2024 - 06.2024 **Scope:** 100&

Project description: SpareBank 1 Midt-Norge (SMN) is one of Norway's largest regional savings banks. The bank needed to make internal routine descriptions more accessible to advisors and employees, and sought a solution capable of providing updated and customised answers to questions based on internal documentation. The project delivered a chatbot based on the RAG technique (Retrieval Augmented Generation), combining large language models with curated internal data. The solution streamlined access to internal knowledge and reduced the time employees spent searching for routines and guidelines.

Role: Data Engineer
Role description: Espen contributed to establishing the data infrastructure for the chatbot solution, including indexing of internal documents in Azure AI Search and storage of data in CosmosDB. He utilised Azure Databricks and Python for data processing and preparation of training data.

Role: GenAI Engineer
Role description: Espen implemented the RAG architecture by combining Azure OpenAI (GPT-4 and GPT-3.5) with indexed internal documents to generate precise and contextually relevant answers. He evaluated and tested Norwegian open language models via Huggingface as alternatives to proprietary models, and contributed to ensuring the quality and relevance of responses for the bank's employees.

Kantega AS
RAG Blueprint

Duration: 03.2024 - 04.2024 **Scope:** 100&
Project description: Kantega is a Norwegian IT consultancy with broad expertise in software development and cloud technology. The company needed to establish best practices for solutions based on RAG architecture (Retrieval Augmented Generation), in order to strengthen its ability to deliver high-quality chatbot and question/answer services to clients. The project delivered a blueprint combining large language models with curated business data, serving as a framework for future client engagements. The solution provided Kantega with a competitive advantage by offering a structured and documented approach to RAG-based solutions.

Role: Developer
Role description: Espen contributed to the development and structuring of the RAG blueprint, implementing solutions that combined large language models with curated business data. He utilised Python and Azure Databricks for the development and testing of RAG pipelines, and ensured the blueprint was well documented and reusable for future projects.

Role: Data Engineer
Role description: Espen established data pipelines for the retrieval, processing, and indexing of business data in Azure AI Search, and utilised CosmosDB for storage of metadata and conversation history. He ensured the data foundation was of high quality and suitable for use in the RAG architecture.

Role: GenAI Engineer
Role description: Espen evaluated and tested both proprietary models such as GPT-4 and open language models via Huggingface, and contributed to defining criteria for model selection and quality assurance of responses in RAG-based solutions.

Elis Norway
Business Intelligence

Duration: 12.2023 - 03.2024 **Scope:** 100&
Project description: Elis Norge was a multiservice provider offering circular rental solutions within mat service, workwear service, and hygiene service. The company was working towards becoming more data-driven, but lacked structured tools for analysis and insight into its own data. The project delivered analytical support and Business Intelligence solutions in Power BI to assist the organisation operationally and support strategic decisions related to establishing a robust data platform. The solution provided Elis Norge with a stronger basis for decision-making and a more solid foundation for its continued digitalisation journey.

Role: Data Analyst
Role description: Espen acted as analyst and advisor, mapping Elis Norge's business needs and data foundation. He analysed existing data to surface operational insights and contributed to identifying which measures and priorities would deliver the greatest value in the company's efforts to become more data-driven.

Role: Power BI Developer
Role description: Espen developed Power BI reports tailored to Elis Norge's business needs, including data modelling and visualisations that gave non-technical users easy access to relevant insights. He contributed

to laying the groundwork for a more structured and scalable approach to reporting and data management within the company.

Norsk Gjeldsinformasjon
Development of Cloud-Based Data Platform

Duration: 01.2023 - 01.2024 **Scope:** 100&
Project description: Norsk Gjeldsinformasjon is a government-appointed entity responsible for managing and sharing debt information in Norway. The company needed a secure and efficient cloud-based data platform to gain better insight into its data, while ensuring compliance with strict privacy requirements and GDPR regulations. The project established a scalable data platform in Azure that provided the organisation with access to structured data analysis and reporting. The solution was handed over with documentation and training to enable the existing team to operate and further develop the platform over time.

Role: Data Engineer
Role description: Espen developed data pipelines and integrations in Azure Synapse Analytics, utilising Python and Kotlin for data processing and transformation. He established structures for secure storage and accessibility of data in accordance with regulatory requirements.

Role: Cloud Architect
Role description: Espen designed the architecture for the cloud-based data platform in Azure, with a focus on security, scalability, and GDPR compliance. He utilised Bicep for infrastructure as code and Azure DevOps for CI/CD, ensuring the platform was robust and maintainable for the internal team.

Role: Power BI Developer
Role description: Espen developed reports and visualisations in Power BI that provided the organisation with insight into its data. He contributed to making insights accessible and understandable for non-technical users within the organisation.

Kantega AS
Data Platform (2023)

Duration: 11.2022 - 01.2023 **Scope:** 100&
Project description: Kantega is a Norwegian IT consultancy with broad expertise in software development and cloud technology. The company needed to map and establish best practices for cloud-based data platforms, in order to strengthen its advisory and delivery capabilities for clients. The project delivered a reference platform built in Microsoft Azure and Databricks, suited for medium to large organisations with requirements for both batch and streaming data processing. The result provided Kantega with a solid foundation for future client engagements within data platforms and cloud architecture.

Role: Data Engineer
Role description: Espen mapped various components in Microsoft Azure and Databricks and developed best practices for cloud-based data platforms. He built a platform supporting both batch and streaming data processing, utilising Bicep for infrastructure as code, Python and Spark notebooks for data processing, and GitHub Actions for CI/CD pipelines. Espen used pytest for automated testing to ensure quality and robustness in the solution.

Role description: Espen contributed to the development of supporting applications around the data platform, utilising bash scripting for process automation and environment setup.

Capgemini
Integration Platform

Duration: 02.2022 - 11.2022 **Scope:** 100&
Project description: Capgemini managed the shared IT systems for pharmacy chains in Norway, responsible for an integration platform handling critical business processes such as inventory management and prescription handling. The platform operated at large scale with high demands for uptime and reliability. The project involved further development and operational support of the platform to ensure stable and

efficient data exchange across the pharmacy chains' systems. The solution contributed to maintaining continuous operations in an industry with strict regulatory requirements and low tolerance for downtime.

Role: Integration developer
Role description: Espen was responsible for the development and operational support of the integration platform, implementing and maintaining integration flows using Aurea Sonic. He utilised XSLT and JavaScript for data mapping and transformation, and Java for business logic. Espen contributed to troubleshooting and support, ensuring high availability in an environment with low tolerance for operational disruptions.

Pasientreiser HF

PRIMO – Design and Development of Pasientreiser's Integration Module

Duration: 09.2021 - 11.2022 **Scope:** 100&
Project description: Pasientreiser HF is a health enterprise owned by the four regional health authorities, responsible for organising and administering patient transport across Norway. The enterprise needed a robust and scalable integration module capable of handling communication and data flow between a range of external actors and systems. The project delivered the PRIMO platform – a central integration solution connecting HelseNorge, Norsk Helsenett, transport providers, map providers, and other external systems. The solution ensured reliable and efficient data exchange, reducing risk associated with gaps in integration within a critical public service function.

Role: Integration developer
Role description: Espen developed and operated integration flows in the PRIMO platform, responsible for the design and implementation of message-based processes between Pasientreiser's systems and external parties. He utilised Aurea Sonic and CXMessenger as the integration platform, and implemented transformations using XSLT and JavaScript. Espen contributed to ensuring high availability and robust error handling in an environment with strict requirements for data quality and uptime within the health sector.

Avella AS

Integration Practice Group

Duration: 08.2021 - 11.2022 **Scope:** 100&
Project description: Avella was a Norwegian IT consultancy specialising in integration and API development. The company needed to establish and maintain a common methodology and toolbox for integration development across platforms and clients, in order to ensure consistent delivery quality and efficient competence building among its consultants. The project delivered structured best practices, competency plans, and course materials for integration development, as well as a professional community ensuring knowledge sharing across teams. The result strengthened Avella's delivery capabilities and helped consultants get up to speed more quickly with new clients and platforms.

Role: Developer and solution architect
Role description: Espen contributed to developing and maintaining Avella's methodology for integration development, with a common approach across platforms. He analysed and tested various integration technologies, prepared competency plans, and contributed to monthly professional gatherings, including hosting a hackathon evening to explore the use of webMethods. He also contributed to standardising the way integrations were built, freeing up time and ensuring consistent delivery quality across clients.

Role: Advisor
Role description: Espen was responsible for Avella Academy within Integration and conducted courses and technical workshops on IBM and Software AG platforms. He was certified in IBM App Connect Enterprise (ACE) and Software AG webMethods, and helped ensure consultants could quickly get up to speed on new platforms and with new clients. He collaborated with other practice groups and served as a subject matter resource across all clients and partners.

Kobler AS

News Trend Analysis

Duration: 04.2019 - 08.2020 **Scope:** 100&

Project description: Kobler was a technology company working to connect advertisers with relevant advertising space in online newspapers. The company needed to identify which news stories and topics were being read unexpectedly often, in order to support the selection of keywords and bids for advertising space in a more data-driven manner. The project delivered an automated solution that analysed reading statistics from online newspapers and extracted defining keywords and trends from the news landscape. The solution gave Kobler a competitive advantage by identifying low-cost advertising space with high exposure, and contributed to driving more targeted sales processes towards clients.

Role description: Espen was solely responsible for the project and developed an MVP from scratch. He utilised Python and related libraries for analysis, and implemented the solution as a Named Entity Recognition (NER) problem to identify relevant keywords – including names of persons, places, and organisations, as well as nouns unique enough to refer to an ongoing news story. Statistical analysis and function approximation were used to assess whether a given keyword constituted an interesting trend. The code was containerised using Docker and deployed to AWS using ECS (Elastic Container Service), making the solution event-based and resource-efficient. Espen was responsible for building the necessary expertise independently and reported periodically to management.

Experience-based qualification

Professional experience

08.2025 - ongoing
PwC Norway

12.2022 - 07.2025
Kantega AS

09.2021 - 11.2022
Avella AS

04.2019 - 06.2020
Kobler AS

Education

2022
Single courses (master level): language technology and machine learning
University of Oslo
Reinforcement Learning and Decision Making Under Uncertainty (10 ECTS): grade A
Adaptive metoder for data-baserte beslutninger (10 ECTS): grade A

08.2018 - 01.2020
Enkeltemner (master): språkteknologi og maskinlæring
University of Oslo

08.2014 - 06.2018
Bachelor Mathematics, Informatics and Technology
University of Oslo

Certifications

01.2024
PL-300 Microsoft Certified: Power BI Data Analyst Associate

02.2022
DP-900 Azure Data Fundamentals

10.2022
Confluent Fundamentals Accreditation

09.2022
IBM Certified Solution Developer – App Connect Enterprise V11

08.2022

05.2022
IBM Event Streams Developer Essentials

09.2021
Certified SAFe® 5 Agilist

Courses

08.2024
Professional Development: Sustainability
Internal course focusing on sustainability in the IT industry, including the environmental impact of technology choices and responsible development of digital solutions.

01.2024
The Consultant School
Internal programme for developing consulting skills, focusing on communication, client management, advisory practices, and professional delivery as an IT consultant.

01.2024
eIDAS 2.0 and Digital Wallets
Internal course on the EU's revised eIDAS regulation and the emergence of digital identity wallets, covering the technical and regulatory implications for digital services.

01.2024
PL-300 Power BI Data Analyst

03.2023
Security by Design
Internal course on integrating security principles early in the development process, focusing on building robust and secure solutions from the ground up rather than adding security as an afterthought.

02.2023
Reliable ML: Applying SRE Principles to ML in Production
Book discussion club on applying Site Reliability Engineering (SRE) principles to machine learning systems in production, focusing on monitoring, reliability, and secure data storage.

11.2022
Store Data in Azure
Self-paced study via Microsoft Learn on storage options in Azure, including relational and non-relational databases, blob storage, and best practices for cloud-based data storage.

11.2022
Automate your workflow with GitHub Actions
Self-paced study via Microsoft Learn on automating development workflows using GitHub Actions, including

setting up CI/CD pipelines and automated build and test processes.

11.2022
Fundamentals of Bicep
Self-paced study via Microsoft Learn on the fundamentals of using Bicep for infrastructure as code in Azure, covering declarative resource definition and automated deployment of cloud infrastructure.

11.2022
Azure for the Data Engineer
Self-paced study via Microsoft Learn aimed at data engineers, focusing on Azure services for data ingestion, storage, transformation, and serving data in the cloud.

06.2022
DevOps: Culture and Practice
Course in DevOps methodology focusing on the cultural and practical aspects of continuous delivery, collaboration between development and operations, and the use of automation to increase delivery speed and quality.

09.2021
Avella Academy
Internal onboarding and training programme at Avella, focusing on integration development, the company's methodology and preferred platforms, and introduction to IBM and Software AG integration tools.

09.2021
Leading SAFe: Scaled Agile Framework

Publications

11.2025
Norway published “The 10 commandments of AI” — here’s what they say, and why they matter
https://medium.com/@espenhk/norway-published-the-10-commandments-of-ai-heres-what-they-say-and-why-they-matter-2b670967eb9b
Norwegian think tank Langsikt has published "The Ten Commandments of AI" – ten guiding principles for Norwegian AI policy, developed by a broad panel of experts from industry, the public sector, and research. The principles span values, foreign policy, infrastructure, research, business, and citizens' rights. A recurring theme is that data should be treated as a public good, that Norway should strengthen its digital sovereignty through European collaboration and open-source alternatives, and that the public sector must build genuine technological competency before deploying AI at scale. The article walks through each commandment and critically evaluates the proposed solutions, with particular emphasis on the view that competency and responsible use are decisive factors in determining whether Norway succeeds.

11.2024
Anonymizing Data in Practice: the why’s, the how’s and the WTF’s [4/4: The Message]
https://medium.com/@espenhk/anonymizing-data-in-practice-the-whys-the-how-s-and-the-wtf-s-4-4-the-message-3fc347ac61b1
Article series on anonymizing data, in 4 parts. Part 4/4.

10.2024
Anonymizing Data in Practice: the why’s, the how’s and the WTF’s [3/4: The WTF]
https://medium.com/kantega/anonymizing-data-in-practice-the-whys-the-how-s-and-the-wtf-s-3-4-75e4fef8f775
Article series on anonymizing data, in 4 parts. Part 3/4.

10.2024
Anonymizing Data in Practice: the why’s, the how’s and the WTF’s [2/4: The How]
https://medium.com/kantega/anonymizing-data-in-practice-the-whys-the-how-s-and-the-wtf-s-2-4-the-how-11237a37771d
Article series on anonymizing data, in 4 parts. Part 2/4.

10.2024
Anonymizing Data in Practice: the why’s, the how’s and the WTF’s [1/4]
https://medium.com/kantega/anonymizing-data-in-practice-the-whys-the-hows-and-the-wtfs-1-4-the-why-ac80fe18f710
Article series on anonymizing data, in 4 parts. Part 1/4.

10.2023
Exploring the Responsible AI Landscape: Books for Every Level of Curiosity and Expertise
https://medium.com/@espenhk/exploring-the-responsible-ai-landscape-books-for-every-level-of-curiosity-and-expertise-fee05a296b8d
Also published: https://www.kantega.no/blogg/exploring-the-responsible-ai-landscape-books-for-every-level-of-curiosity-and-expertise

06.2022
Breaking down AlphaGo - Reinforcement Learning in Games
https://www.youtube.com/watch?v=ZclWqNVEegw
This video presentation breaks down Google DeepMind's "AlphaGo" paper, which documented the release of a reinforcement learning-based model that was the first AI to surpass top human performance, demonstrated by its convincing victory over world champion Lee Sedol. The presentation was created as part of the course IN-STK5100 "Reinforcement Learning" at the University of Oslo in spring 2022, and received a grade of A.

08.2024
Reproducible analysis of limited datasets, project 1 (IN-STK5000 exam project)
https://www.youtube.com/watch?v=mrdj9-g1MX4
This project is an exam project for the course IN-STK5000 at the University of Oslo: "Responsible Data Science" (formerly "Adaptive methods for data-based decision making").

The project is tasked with training an ML classifier to predict whether a patient should be checked for diabetes, using a very limited dataset of 500 individuals with limited data quality. The focus was on implementing a reproducible approach that made it simple to re-run, adapt and evaluate experiments, also for people outside the group. Our assigment was graded to an A, 40 of 40 marks.

Repo for code: https://github.com/YSLCoat/IN-STK5000/tree/main/oblig2

10.2024
Reproducible analysis of limited datasets, project 2 (IN-STK5000 exam project)
https://www.youtube.com/watch?v=pXGPLKWdVls
This project is an exam project for the course IN-STK5000 at the University of Oslo: "Responsible Data Science" (formerly "Adaptive methods for data-based decision making").

The project is tasked with training an ML classifier to predict whether a patient should be checked for diabetes, using a very limited dataset of 500 individuals with limited data quality. The focus was on implementing a reproducible approach that made it simple to re-run, adapt and evaluate experiments, also for people outside the group. Our assigment was graded to an A, 40 of 40 marks.

Repo for code: https://github.com/YSLCoat/IN-STK5000/tree/main/oblig2

Languages

Norwegian
Fluent

English
Basic

German
Basic

Italian
Basic